Billal Zughbi

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PROFESSIONAL SUMMARY

Business Intelligence and Analytics Engineer with 3.5 years of experience building enterprise reporting and data solutions across an 8-business-unit manufacturing organization. Develops Power BI and SQL solutions that form business insights, replace manual processes, standardize reporting, and connect enterprise data at scale. Hands-on experience with Snowflake, Microsoft Fabric, Python, data modeling, and ETL/ELT.

WORK EXPERIENCE

Analytics Engineer | Arconic — Lancaster, PA *January 2023 – Present*

- Designed and deployed 15+ enterprise Power BI dashboards for 200+ stakeholders across operations, finance, and executive leadership, replacing legacy spreadsheet reporting with a centralized reporting layer for core KPIs.
- Analyzed manufacturing and product-quality data to identify key drivers of quality failures, informing process improvements that reduced defect rates by 25% and annual customer claims by $80K.
- Built a SQL data model consolidating $12M in capital spend and forecast data, improving visibility into departmental spending and identifying $200K+ in potentially redundant expenditures.
- Automated resource-planning workflows in SQL and Power BI by translating business rules and operational constraints into reporting logic, eliminating approximately 90 hours of manual scheduling each month.
- Standardized ETL workflows across 8 business units by establishing common naming conventions, schema definitions, and transformation logic across previously unit-specific pipelines.
- Integrated Snowflake and Microsoft Fabric lakehouse data into Power BI reporting models, reducing direct queries against legacy IBM AS/400 and Microsoft Access systems for enterprise reporting.

Data Analytics Intern | Arconic — Lancaster, PA *May 2022 – August 2022*

- Automated the collection and transformation of legacy customer-service data using Power Automate, creating a refreshable reporting dataset that reduced monthly preprocessing time by 40%.
- Built a Power BI and Power Apps attendance-tracking platform for 750+ hourly employees, standardizing data capture and KPI logic while reducing reliance on manual timecard audits.
- Developed technical documentation and user guides for analytics tools, creating standardized resources for onboarding and troubleshooting across citizen development teams.

EDUCATION

The College of New Jersey — Ewing, NJ *December 2022*

Bachelor of Science in Computer Science

TECHNICAL PROJECTS

Apple Health Analytics Dashboard | GitHub Repository *March 2025, Rebuilt October 2026*

- Built a Python pipeline using ElementTree and Pandas to extract, de-duplicate, and normalize 4.3M+ records from Apple Health XML data across 30 health metrics, spanning 12 years and multiple devices.
- Modeled the data in Power BI using a relational schema and custom DAX measures, then audited all 24 KPI cards in Pandas and rebuilt the report's 4 pages for the web.

SKILLS

Business Intelligence: Power BI, DAX, Power Query (M), Excel, Row-Level Security (RLS)

Data & Engineering: SQL, Snowflake, Microsoft Fabric, SQL Server, Python, ETL/ELT, Data Modeling, Star Schema

Automation & Development: Power Automate, Power Apps, Git, Azure DevOps